

352 PCS Dataset - Pitch-Class Sets Sorted by Roughness

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Version: 1.0

Publication date: 7 October 2026

DOI: <https://doi.org/10.5281/zenodo.23214708>

Description

The **352 PCS Dataset** contains 352 pitch-class sets represented as structured records and ordered according to roughness-related measures.

The dataset was generated using a spectral model based on:

- **Number of harmonics:** 32
- **Base note:** MIDI 60

Each pitch-class set is represented by an integer identifier and an associative record containing pitch content, roughness measures, interval-class information, and parent/child relationships between sets.

The dataset is intended for research in computational music theory, psychoacoustics, algorithmic composition, electroacoustic music, and music computing.

Dataset structure

Each record contains the following fields:

- **integerNumber** - integer representation of the pitch-class set
- **chord** - pitch-class content
- **rough** - calculated roughness value
- **numIntervalliChord** - number of intervals considered in the chord
- **roughMedia** - mean roughness
- **vettoreIntervallare** - interval-class vector
- **roughVI** - roughness associated with the interval-class vector
- **vettoreIntervallareConUnisono** - interval-class vector including unisons
- **roughVIU** - roughness associated with the interval-class vector including unisons
- **figliChord** - child pitch-class sets
- **figliIntegerNumber** - integer identifiers of child sets
- **padiIntegerNumber** - integer identifiers of parent sets

- `padriChord` - parent pitch-class sets

Files

The repository contains the dataset and its derived representations in several formats:

- `scrittura01 Associazioni Roughness Dataset.nb`
Wolfram Mathematica notebook containing the dataset.
- `scrittura01 Associazioni Roughness Dataset&Code.nb`
Wolfram Mathematica notebook containing the dataset and related source code.
- `352_PCS_Dataset_Roughness_H32_MIDI60.json`
Machine-readable JSON representation of the complete dataset.
- `352_PCS_Dataset_Roughness_H32_MIDI60.pdf`
Human-readable PDF representation of the dataset and graphical outputs.
- `352_PCS_Dataset_Source.zip`
Markdown and LaTeX source files together with the `images` directory required to preserve relative figure links.
- `2026_10_07_15_24_59_360midiFile.mid`
MIDI material associated with the dataset.

JSON format

The JSON file provides a software-independent representation of the dataset.

The original ordering of the Wolfram Association is preserved explicitly in the `order` array. Keys in the `dataSet` object are JSON strings corresponding to the original integer keys.

This format allows the dataset to be processed without requiring Wolfram Mathematica.

Figures and document sources

The Markdown and LaTeX documents reference graphical files using relative paths to the `images` directory.

When using the source package, keep the following structure unchanged:

```
352_PCS_Dataset_Source/
|-- 352_PCS_Dataset_Roughness_H32_MIDI60.md
|-- 352_PCS_Dataset_Roughness_H32_MIDI60.tex
`-- images/
    `-- ...
```

Repository

GitHub:

<https://github.com/FrancescoScagliola/Repository/tree/main/352%20PCS%20Dataset>

DOI

Zenodo:

<https://doi.org/10.5281/zenodo.23214708>

Citation

If you use this dataset in research, please cite:

Scagliola, F. (2026). *352 PCS Dataset: Roughness Measures and Interval-Class Associations* (Version 1.0) [Dataset]. Zenodo.
<https://doi.org/10.5281/zenodo.23214708>

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See:

`LICENSE-DATA.txt`

Source code is licensed under the **GNU General Public License v3.0 (GPL-3.0)**.

See:

`LICENSE-CODE.txt`

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